NUCLEUS

Centre of Excellence

JEE ADVANCED PATTERN

TEST-11 (29-09-2021)

Time: 3 Hours | OPEN TEST SERIES | **Maximum Marks: 198**

PAPER-1 WITH SOLUTION

READ THE INSTRUCTIONS CAREFULLY

GENERAL

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. The paper CODE is printed on the right hand top corner of this sheet and the right hand top corner of the back cover of this booklet.
3. Use the Optical Response Sheet (ORS) provided separately for answering the questions.
4. The paper CODE is printed on the left part as well as the right part of the ORS. Ensure that both these codes are identical and same as that on the question paper booklet. If not, contact the invigilator for change of ORS.
5. Blank spaces are provided within this booklet for rough work.
6. Write your name, roll number and sign in the space provided on the back cover of this booklet.
7. After breaking the seal of the booklet verify that the booklet contains **27 pages** and that all the **54 questions** along with the options are legible. If not, contact the invigilator for replacement of the booklet.
8. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET

9. Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's Sheet.
10. The ORS will be collected by the invigilator at the end of the examination.
11. You will be allowed to take away the Candidate's Sheet at the end of the examination.
12. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
13. Write your name, roll number and code of the examination center, and sign with pen is the space provided for this purpose on the ORS. Do not write any of these details anywhere else on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

14. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
15. Darken the bubble ◯ **COMPLETELY**.
16. The correct way of darkening a bubble is as: ⬤
17. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
18. Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

Please see the last page of this booklet for rest of the instructions.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR.

Name of the Candidate

I have read all the instructions and shall abide by them.

Signature of the Candidate

Form Number

I have verified all the information filled in by the Candidate.

Signature of the invigilator

CHOOSE WISE TO RISE

Some Useful Data

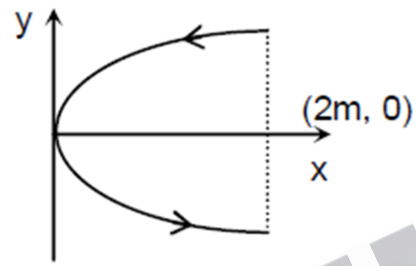
Quantity	Values
Constant of gravitation	6.67259×10^{-11} N-m^2 kg^{-2}
Speed of light in vacuum	2.99792458×10^8 m s^{-1}
Avogadro constant	6.0221367×10^{23} mol^{-1}
Gas constant	8.314510 JK^{-1}-mol^{-1}
Boltzmann constant	1.380658×10^{-23} JK^{-1}
	8.617385×10^{-5} eV K^{-1}
Stefan-Boltzmann constant	5.67051×10^{-8} W m^{-2} - K^{-4}
Wien's displacement law constant	2.897756×10^{-3} m -K
Charge of proton	$1.60217733 \times 10^{-19}$ C
Mass of electron	$9.1093897 \times 10^{-31}$ kg
	$5.48579903 \times 10^{-4}$ u
Mass of proton	$1.6726231 \times 10^{-27}$ kg
	1.007276470 u
Permeability of vacuum	$4\pi \times 10^{-7}$ NA^{-2}
Permittivity of vacuum	$8.854187817 \times 10^{-12}$ C^2 N^{-1} m^{-2}
Faraday constant	96485.3029 C mol^{-1}
Planck constant	$6.6260755 \times 10^{-34}$ J-s
	$4.1356692 \times 10^{-15}$ eV-s

PART-I : PHYSICS

SECTION-I : Single Correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. You will be awarded **3 marks** if only the correct option is chosen and zero mark if none of the option is chosen. **(–1)** marks will be awarded for incorrect answers in this section.

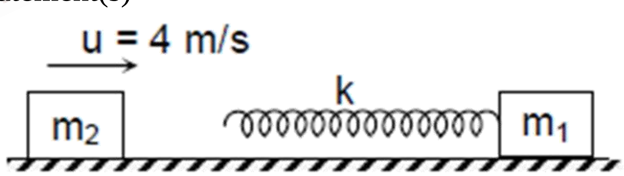
1. A conducting wire is bent in the form of a parabola $y^2 = 2x$ carrying a current $i = 2A$ as shown in the figure. The wire is placed in a uniform magnetic field $\vec{B} = -4\hat{K}$. The magnetic force on the wire is



(A) $-16\hat{i}$ (B) $32\hat{i}$ (C) $-32\hat{i}$ (D) $16\hat{i}$

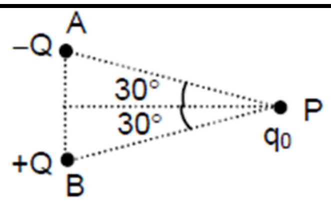
2. If M = mass, L = length, T = time and I = electric current, then the dimensional formula of resistance R will be given by
(A) $[R] = [M^1L^2T^{-3}I^{-2}]$ (B) $[R] = [M^1L^2T^{-3}I^2]$ (C) $[R] = [M^1L^2T^3I^{-2}]$ (D) $[R] = [M^1L^2T^3I^2]$

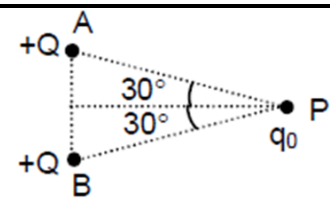
3. A block of mass = 3 kg is connected with ideal spring of spring constant k = 30000 N/m and kept at horizontal frictionless surface as shown in the figure. The block m_2 = 1 kg is moving with velocity u = 4 m/s towards block m_1. v_1 and v_2 are the final speed of the block m_1 and m_2 respectively.
Choose the correct statement(s)



(A) $v_1 = v_2$ = 1m/s
(B) The maximum compression in the spring is 2 cm
(C) The maximum compression in the spring is 4 cm
(D) none

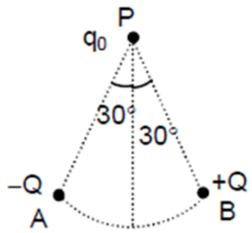
4. The diagram shows four systems of charged particles. Each system contains two charged particles at point A and point B as shown in the figure. A third positive point charge q_0 is kept at point P in each system. The direction of electrostatic force experienced by the charge q_0 will be

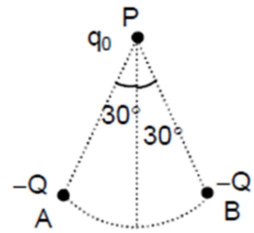




System –1; AP = BP = r and AB is a straight line

system –2: AP = BP = r and AB is a straight line



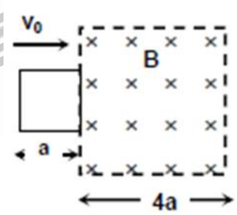


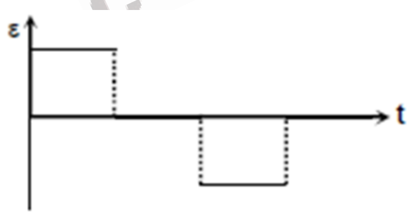
System -3; AP = BP = r and AB is a circular arc of radius r and centre at O

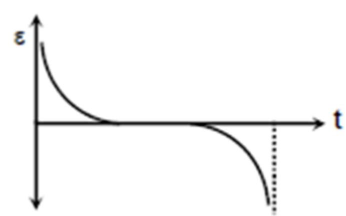
System -4; AP = BP = r and AB is a circular arc of radius r and centre at O

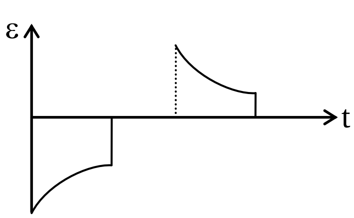
(A) **System-1** $\uparrow$ **System-2** $\rightarrow$ **System-3** $\leftarrow$ **System-4** $\downarrow$
(B) **System-1** $\downarrow$ **System-2** $\leftarrow$ **System-3** $\leftarrow$ **System-4** $\uparrow$
(C) **System-1** $\uparrow$ **System-2** $\rightarrow$ **System-3** $\rightarrow$ **System-4** $\downarrow$
(D) **System-1** $\uparrow$ **System-2** $\rightarrow$ **System-3** $\leftarrow$ **System-4** $\uparrow$

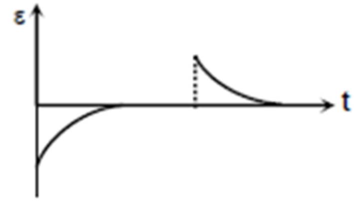
5. A square loop of side a is projected at t = 0, with speed v_0 into a region having uniform magnetic field B, with velocity perpendicular to the magnetic field. Which of the following plots correctly represents the variation of induced emf of loop with time



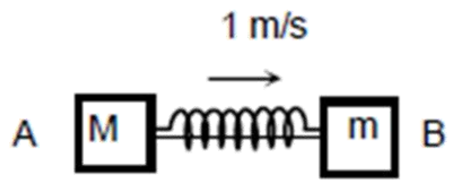
(A)


(B)


(C)


(D)


6. A block A of mass M = 2kg is connected to another block B of mass 1kg with a string and a spring of force constant k = 600 N/m as shown in the figure. Initially spring is compressed to 10 cm and whole system is moving on a smooth surface with a velocity v = 1 m/s. At any time thread is burnt, the velocity of block A, when B is having maximum velocity w.r.t. ground, is:



(A) 0 (B) 1 m/s (C) 3 m/s (D) none of these

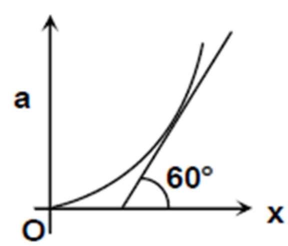
SECTION-II : One or More Options Correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE THAN ONE** may be correct. You will be awarded **4 marks** if all the correct option(s) is/are chosen and zero mark if none of the option is chosen. For each correct answer **+1** mark will be awarded if no incorrect option is chosen. In all other cases minus two **(–2)** mark will be awarded.

7. In a YDSE set up the separation between the slits is 3×10^{-3} m. The distance between the slits and the screen is 1.5 m. Light of wavelength in the visible range (4000Å - 8000 Å) is allowed to fall on the slits. The wave length in the visible region that will be present on the screen at a distance of 1.5×10^{-3} m from central maxima is

(A) 4285.7Å (B) 5000 Å (C) 6000 Å (D) 7500 Å

8. A particle starts moving with initial velocity 3 m/s along x-axis from origin. Its acceleration is varying with x in parabolic nature as shown in figure. At x = $\sqrt{3}$ m tangent to the graph makes an angle 60° with positive x-axis as shown in diagram. Then at x = $\sqrt{3}$



(A) $v = \sqrt{\left(\sqrt{3} + a\right)}$ m/s (B) a = 1.5 m/s^2

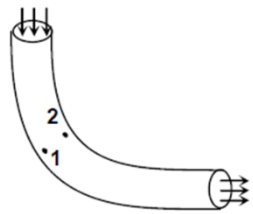
(C) v = $\sqrt{12}$ m/s (D) a = 3 m/s^2

9. A vertical capillary tube with inside radius 0.25 mm is submerged into water so that the length of its part protruding over the water surface is equal to 25 mm. surface tension of water is 73×10^{-3} N/m and angle of contact is zero degree for glass and water, acceleration due to gravity is 9.8 m/s^2. Then choose correct statement.

(A) R = 0.25 mm (B) h = 59.6mm (C) R = 0.60 mm (D) h = 25 mm

where R is radius of meniscus and h is height of water in capillary tube.

10. Ideal fluid flows along a tube of uniform cross section, located in a horizontal plane and bent as shown in figure. The flow is steady, 1 and 2 are two points in the tube. If P_1 and P_2 are pressure at the two points and v_1 and v_2 are the respective velocities, then



(A) $P_1 < P_2$ (B) $P_1 > P_2$ (C) $v_1 < v_2$ (D) $v_1 > v_2$

11. In the series L – C – R circuit, the voltage across resistance, capacitance and inductance are 30V each at frequency $f = f_0$.

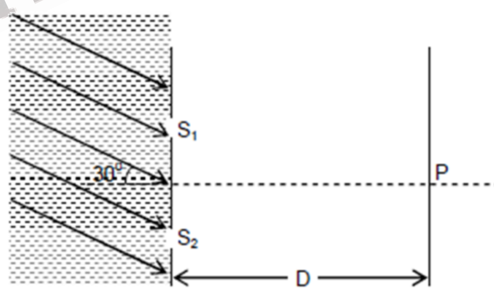
(A) It the inductor is short-circuited, the voltage across the capacitor will be $30\sqrt{2}$ V.

(B) If the capacitor is short-circuited, the voltage drop across the inductor will be $\frac{30}{\sqrt{2}}$ V.

(C) If the frequency is changed to $2f_0$, the ratio of reactance of the inductor to that of the capacitor is 4 : 1.

(D) If the frequency is changed to $2f_0$, the ratio of the reactance of the inductor to that of the capacitor is 1 : 4.

12. The given figure shows a YDSE apparatus. Parallel monochromatic coherent light rays are incident on slits S_1 and S_2 ($S_1S_2 = \frac{2}{3}$ mm) at an angle 30° with the horizontal. The medium on left side of the slits is water ($\mu_g = 4/3$). To obtain the central maxima at point P, a glass slab ($\mu_g = 3/2$) inside water is introduced in front of slit S_1. The thickness of the glass slab required for this purpose is

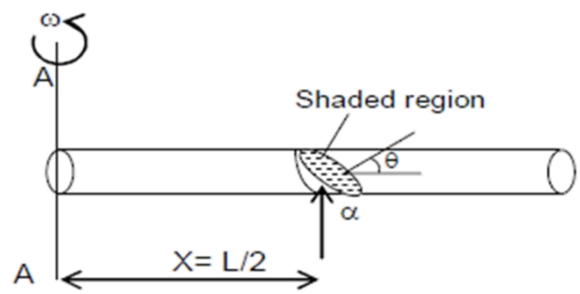


(A) 2mm (B) 4/3mm (C) 4 mm (D) 8/3 mm

SECTION-III : Integer Value Correct Type

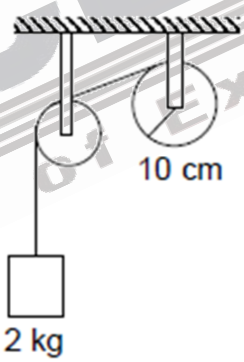
This section contains **6 questions**. For each question, enter the correct numerical value (in decimal notation, truncated / rounded off to the **second decimal place;** e.g. 6.25, 7.00, –0.33, –0.30, 30.27, –127.30) using the mouse and the on screen virtual numeric keypad in the place designated to enter the answer. You will be awarded **4 marks** if correct numerical value is entered as answer. **No negative marks** will be awarded for incorrect answers in this section.

13. A thin uniform metallic rod of mass m, length L, Young modulus of elasticity Y and cross-sectional area A is rotated by angular velocity ω about extreme end about axis AA'. Consider a section on the rod at midpoint of rod. What will be the normal stress in N/m^2 on the shaded region? (Take the value of $\frac{M\omega^2L^2\cos^2\theta}{A} = 10N/m^2$)

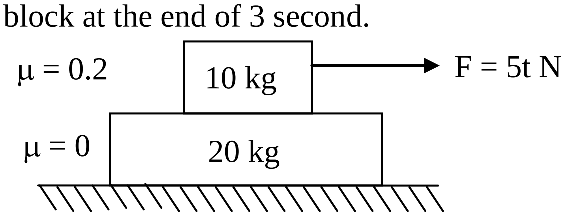


14. The thickness of a glass plate is measured to be 2.17 mm, 2.17 mm and 2.18 mm at three different places. Find the average thickness of the plate from this data, (in mm).

15. A string is wrapped on a wheel of moment of inertia 0.20 kg m^2 and radius 10 cm and goes through a light pulley to support a block of mass 2.0 kg as shown in figure. Find the acceleration of the block. (in m/s^2)

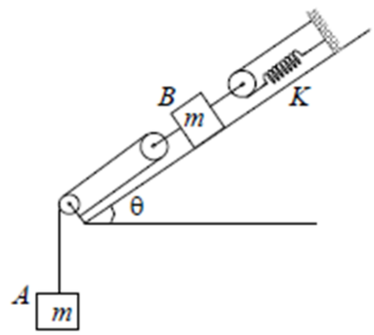


16. Find the speed of 10kg block at the end of 3 second.



17. A uniform disc of radius R having change Q distributed uniformly all over its surface is placed on a smooth horizontal surface. A magnetic field, $B = kxt^2$, where k is a constant, x is the distance (in metre) from the centre of the disc and t is the time (in second), is switched on perpendicular to the plane of the disc Find the torque (in N-m) acting on the disc after 15 sec. (Take 4KQ = 2.50 S.I. unit and R = 1m).

18. Two block A and B, each of mass m are connected by means of a pulley-spring system on a smooth inclined plane of inclination θ as shown in the figure. All the pulleys and spring are ideal. Now, B is slightly displaced from its equilibrium position. It starts to oscillate. Time period of oscillation of B will be (Take m = 4kg, k = 5 N/m, π = 3.14)



PART-II : MATHSMATICS

SECTION-I : Single Correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. You will be awarded **3 marks** if only the correct option is chosen and zero mark if none of the option is chosen. **(–1)** marks will be awarded for incorrect answers in this section.

19. There are 7 distinguishable rings. The number of possible five-ring arrangements on the four fingers (except the thumb) of one hand (the order of the rings on each finger is to be counted and it is not required that each finger has a ring) is

(A) 214110 (B) 211410 (C) 124110 (D) 141120

20. If the equation of the plane passing through the intersection of the planes $x + 2y + z - 1 = 0$ and $2x + y + 3z - 2 = 0$ and perpendicular to the plane $x + y + z - 1 = 0$ is $x + ky + 3z - 1 = 0$, then the value of k is

(A) 4 (B) – 4 (C) 2 (D) – 2

21. If $f(x) = (m + 2)x^3 - 3mx^2 + 9mx - 1$ decreases on R, then m belongs to

(A) $(-\infty, -3) \cup (0, \infty)$ (B) $(-\infty, -3]$
(C) $(-\infty, -3)$ (D) $[-1, 0] \cup (0, 1]$

22. If curve C_1 is image of curve $C_2 : 7x^2 - 9y^2 = 63$ in the line $x + y = \sqrt{2}$, then equation of common normal to both the curves at their common point is -

(A) $x - y = 16\sqrt{2}$ (B) $x - y = 4\sqrt{2}$ (C) $x - y = 8\sqrt{2}$ (D) $x - y = 64\sqrt{2}$

23. A bag contains 2 different red, 4 different white and 4 different blue balls. Three balls are drawn one by one at random without replacement. The probability that first and last draw results in different colour balls, is

(A) $\frac{22}{45}$ (B) $\frac{28}{45}$ (C) $\frac{31}{45}$ (D) $\frac{32}{45}$

24. The coefficient of x^5 in the expansion of $(1 + x^2)^3 (1 + x^4)^7 (1 - x)^3$ is

(A) –30 (B) –33 (C) –21 (D) –12

SECTION-II : One or More Options Correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE THAN ONE** may be correct. You will be awarded **4 marks** if all the correct option(s) is/are chosen and zero mark if none of the option is chosen. For each correct answer **+1** mark will be awarded if no incorrect option is chosen. In all other cases minus two **(–2)** mark will be awarded.

25. If $1, z_1, z_2, ... z_{10}$ are the roots of equation $z^{11} - 1 = 0$, then -
(where $i = \sqrt{-1}$ and $\omega(\neq 1)$ is complex cube root of unity)

(A) $\prod_{k=1}^{10}(i - z_k) = i$ (B) $\prod_{k=1}^{10}(i - z_k) = -i$ (C) $\prod_{k=1}^{10}(\omega + z_k) = \omega$ (D) $\prod_{k=1}^{10}(\omega + z_k) = \omega^2$

26. Let $P(x_0, y_0)$ be any point on the parabola $y^2 = 4x$, if normal at P intersect the parabola again at Q and tangents at P and Q intersect at point R -

(A) acute angle between normal at P and tangent at Q to the parabola is θ, then $\tan\theta = \frac{|y_0|}{4}$

(B) length of normal chord PQ is $\frac{4(x_0+1)^{3/2}}{x_0}$

(C) ordinate of point R is $-\frac{4}{y_0}$

(D) ordinate of circumcentre of triangle PQR is $\frac{-2(x_0+3)}{y_0}$

27. If $y = y(x)$, $y(x) \in \left[0, \frac{\pi}{2}\right)$ is solution of $\left[\frac{1+2\sin x\cos^2(x+y)}{\cos^2(x+y)}\right]dx + \left[\frac{1}{\cos^2(x+y)} + 2\sec^2 y\right]dy = 0$,

$y(0) = \frac{\pi}{4}$, then which is/are true -

(A) $y\left(\frac{\pi}{2}\right) = \frac{\pi}{4}$

(B) $y\left(\frac{\pi}{2}\right) = -\frac{\pi}{4}$

(C) Equation of tangent to $y = y(x)$ at $\left(\frac{\pi}{2}, \frac{\pi}{4}\right)$ is $8x + 12y = 7\pi$

(D) Equation of tangent to $y = y(x)$ at $\left(\frac{\pi}{2}, \frac{\pi}{4}\right)$ is $12x + 8y = 7\pi$

28. Let S be the set of positive value(s) of r such that the area enclosed by the tangents drawn from the point P(6, 8) to the circle $x^2 + y^2 = r^2$ and the chord of contact is $\frac{75\sqrt{3}}{4}$ sq. units. Then the correct option(s) is/are -

(A) S is singleton set

(B) Equation of chord of contact is $6x + 8y = \alpha^2$, $\alpha \in S$

(C) Let A and B are point of contact of tangents then the equation of circumcircle of ΔABP is $x^2 + y^2 - 6x - 8y = 0$

(D) Number of elements in set S is 2

29. A continuous function f : R → R is defined as $f(x) = \begin{cases} \dfrac{\sin(p+q)x + \sin 2x}{x} & , \quad x < 0 \\ q & , \quad x = 0 \\ \dfrac{(x+rx^2)^{1/2} - \sqrt{x}}{rx\sqrt{x}} & , \quad x > 0 \end{cases}$, then which of the following is(are) correct?

(A) $q = \dfrac{1}{2}$

(B) p = –2

(C) p + q + r = $-\dfrac{3}{2}$

(D) Complete set of values of r is R – {0}

30. The sum of the third and ninth terms of an AP is equal to the least value of the quadratic expression $2x^2 - 4x + 10$. Then

(A) the common difference of the AP is $\dfrac{4-a}{5}$ where "a" is the first term of the AP.

(B) if the first term a is –1, then the common difference is 1

(C) the common difference d cannot be determined unless the first term is given

(D) sum of the first eleven terms is 44

SECTION-III : Integer Value Correct Type

This section contains **6 questions**. For each question, enter the correct numerical value (in decimal notation, truncated / rounded off to the **second decimal place;** e.g. 6.25, 7.00, –0.33, –0.30, 30.27, –127.30) using the mouse and the on screen virtual numeric keypad in the place designated to enter the answer. You will be awarded **4 marks** if correct numerical value is entered as answer. **No negative marks** will be awarded for incorrect answers in this section.

31. Given a = cosθ + isinθ and the equation $az^2 + z + 1 = 0$ has a purely imaginary root and $f(x) = x^3 - 3x^2 + 3(1 + \cos\theta)x + 5$. If

p = Number of points of local extrema of y = f(x)

q = Number of points of inflection of y = f(x)

r = Number of points of negative real roots of equation f(x) = 0, then 'p + q + r' is equal to

32. If f is a real-valued function defined for all x ≠ 0, 1 and satisfying the relation

$$f(x) + f\left(\frac{1}{1-x}\right) = \frac{2}{x} - \frac{2}{1-x}$$

Then $2019 \lim\limits_{x \to 2020} f(x)$ will be

33. Let S be the set of all symmetric matrices of order 3×3, all of whose elements are either 0 or 1. If five of these elements are 1 and four of them are 0, then the number of matrices in S is

34. If the equation of the curve obtained after reflection about the line x – y = 2 of the ellipse $\frac{(x-4)^2}{16}+\frac{(y-3)^2}{9}=1$, is $16x^2 + 9y^2 + k_1x - 36y + k_2 = 0$. Then value of $(k_1 + k_2)$ will be

35. Derivative of $\ln^2(\ln x)$ with respect to $\ln(\ln x)$ at $x = e^e$ (where e is Napier's constant)

36. Let $L_1 : \frac{x-1}{3}=\frac{y-2}{1}=\frac{z-3}{-3}$ be a line and P : 4x + 3y + 5z = 50 be a plane. L_2 is the line in the plane P and parallel to L_1. If the equation of plane containing both the lines L_1 and L_2 and perpendicular to plane P is $14x - by + 5z + d = 0$ (b, d $\in$ R), then (b + d) is equal to

PART-III : CHEMISTRY

SECTION-I : Single Correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. You will be awarded **3 marks** if only the correct option is chosen and zero mark if none of the option is chosen. **(–1)** marks will be awarded for incorrect answers in this section.

37. The CFSE for $[CoCl_6]^{4-}$ complex is 18000 cm^{-1}. The Δ for $[CoCl_4]^{2-}$ will be

(A) 9000 cm^{-1} (B) 4000 cm^{-1} (C) 8000 cm^{-1} (D) 2000 cm^{-1}

38. Which of the following statement for crystal having schottky defect is incorrect.

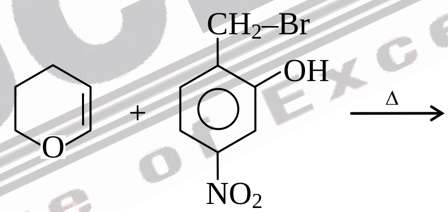
(A) Schottky defect arises due to absence of cations& anion from positions which they are expected to occupy.

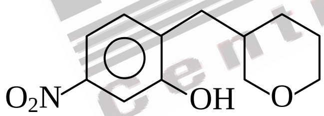
(B) The density of crystal having shottky defect is smaller than that of perfect crystal.

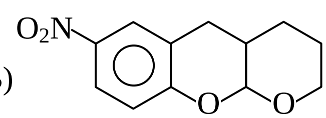
(C) Schottky defect are more common in ionic compounds with higher co-ordination number.

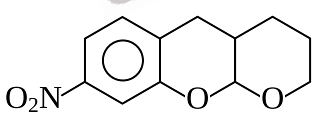
(D) Shottky defect is a non-stoichiometric defect.

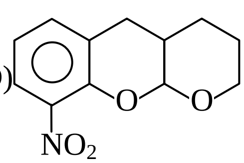
39. Major product of the following reaction is :



(A) 

(B) 

(C) 

(D) 

40. Which of the following ore is oxide ore ?

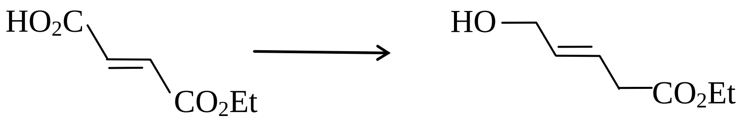
(A) Magnesite (B) Azurite (C) Ilmenite (D) Cryolite

41. For a zero order reaction : $A \longrightarrow 2B$

Initial rate of reaction is 0.1M/min. If concentration of 'A' is 0.1M after 120 sec then what would be concentration of B after 60 sec.

(A) 0.2 M (B) 0.1 M (C) 0.05 M (D) 0.15 M

42. Which of the following can be used for given transformation?



(A) $NaBH_4$/MeOH (B) DIBAL, –78°C
(C) B_2H_6.THF (D) H_2/ Raney Ni

SECTION-II : One or More Options Correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE THAN ONE** may be correct. You will be awarded **4 marks** if all the correct option(s) is/are chosen and zero mark if none of the option is chosen. For each correct answer **+1** mark will be awarded if no incorrect option is chosen. In all other cases minus two **(–2)** mark will be awarded.

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43. If z-axis be the internuclear axis, then bond would be formed by the overlap between:

(A) s and p_z (B) p_z and p_z (C) p_y and p_y (D) d_{z^2} and d_{z^2}

44. Select the **incorrect** option(s).

(A) Larger the value of T_C easier is the liquification of gas.

(B) At given temperature, a gas with low molar mass has less translational kinetic energy than a gas with high molar mass.

(C) Molar volume of a gas $V_m > 22.4$ L at 1 atm and 273 K when attractive force dominates between gas particles.

(D) The real gas behaves like an ideal gas at Boyle's temperature & at all pressures.

45. If R shows aldol reaction and T shows disproportionation reaction then which of the following statements are correct :

$$\text{Propene} \xrightarrow[\text{(ii) } Ph_3P \text{ (iii) LDA}]{\text{(i) NBS / } CCl_4/\Delta} (P) \xrightarrow{\text{Acetaldehyde}/\Delta} (Q) \xrightarrow[\text{Zn/ } H_2O]{O_3} (R) + (S) + (T)$$

(A) Q produces same major product with HBr (1 equivalent) at 80 °C & – 80°C.

(B) All R, S & T produces silver mirror with ammoniacal $AgNO_3$.

(C) Q on reaction with H_2/Ni (1 equivalent) produces cis-2-butene.

(D) Baeyer's reagent can produce acetic acid on reaction with compound R.

46. Which of the following has tetrahedral shape with d^3s Hybridization ?

(A) $KMnO_4$ (B) K_2CrO_4 (C) $KClO_4$ (D) NH_3

47. An oleum sample has SO_3 and H_2SO_4 in 2 : 3 mass ratio. Select the correct statement(s) about this sample :

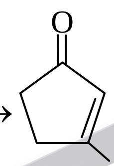
(A) Percentage strength of oleum sample is 109%

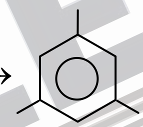
(B) Percentage strength of oleum sample is 118%

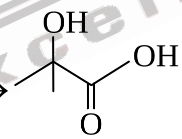
(C) Percentage of free SO_3 in the sample is 40 %

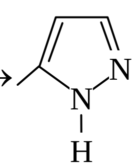
(D)18 g of water is required to convert all the free SO_3into H_2SO_4, present in 200 g of oleum sample

48. Which of the following are correctly matched with major product?

(A) (2R, 5S)-Hex-3-yn-2,5-diol $\xrightarrow[\text{(ii) } N_2H_4/H_2O_2 \text{ (iii) } CrO_3 \text{ (iv) NaOH, } \Delta]{\text{(i) } H_2\text{, Pd/BaSO}_4}$ 

(B) (S)-5-Methyl hex-4-en-2-ol $\xrightarrow[\text{(ii) Conc. } H_2SO_4\text{, }\Delta]{\text{(i) } KMnO_4\text{, }\Delta}$ 

(C) (2R, 5R)-3,4-Dimethyl hex-3-endiol $\xrightarrow[\text{(ii) } Ag(NH_3)_3^+ \text{ (iii) NaOH, } \Delta]{\text{(i) } O_3/H_2O_2}$ 

(D) (R)-(E)-3-Methyl hept-3-en-1,6-diol $\xrightarrow[\text{(ii) } O_3\text{, Zn/}H_2O \text{ (iii) } N_2H_4 \text{(1eq.)}]{\text{(i) PCC}}$ 

SECTION-III : Integer Value Correct Type

This section contains **6 questions**. For each question, enter the correct numerical value (in decimal notation, truncated / rounded off to the **second decimal place;** e.g. 6.25, 7.00, –0.33, –0.30, 30.27, –127.30) using the mouse and the on screen virtual numeric keypad in the place designated to enter the answer. You will be awarded **4 marks** if correct numerical value is entered as answer. **No negative marks** will be awarded for incorrect answers in this section.

49. $S_2O_3^{2\ominus} + I_2 \longrightarrow I^{\ominus} + X \text{ (ion)}$

What is the average oxidation state of central atom of X ion.

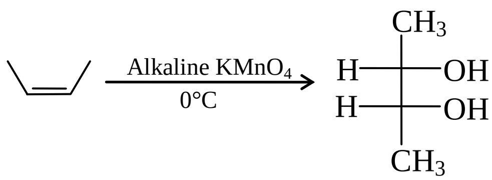
[Multiply your answer by 100]

50. A mixture of NH_3 (g) and N_2H_4 (g) is placed in a sealed rigid container at 300 K. Total pressure of the gas mixture is 0.5 atm. Now container is heated to 1200 K, where following complete decomposition reactions take place :

$$2NH_3 \rightarrow N_2 + 3H_2$$
$$N_2H_4 \rightarrow N_2 + 2H_2$$

Pressure of the vessel at this stage becomes 4.5 atm. If the mole fraction of N_2H_4 (g) in the original mixture is X, then find the value of 1/X.

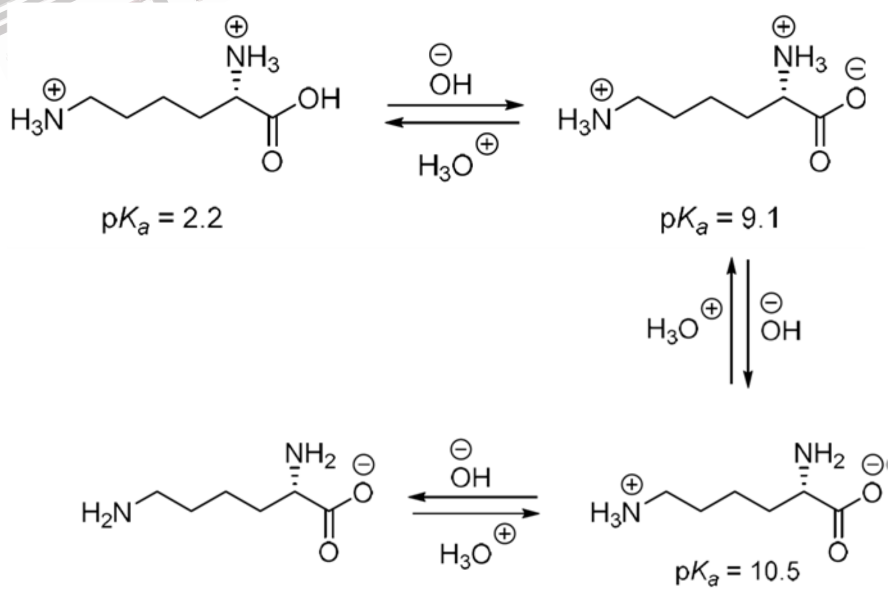
51. For the following redox reaction, how many moles of $KMnO_4$ required for 12 moles of given alkene.



52. Consider the following complexes.
(i) $[Cr(EDTA)]^-$ (ii) $[Ru(en)_3]^{2+}$
(iii) $[Pt(gly)Cl_2]^-$ (iv) Cis-$[Cr(Cl_2)(ox)_2]^{3-}$
(v) trans-$[Pt(en)(NH_3)_2(py)_2]^{4+}$
Then find out the value of expression $|p \times q|^2$.
p = Total number of optically active complexes
q = Total number of optically inactive complexes

53. From 196 mg of H_2SO_4, 6.023×10^{20} molecules were removed and remaining H_2SO_4 was neutralised by 20 ml NaOH. If molarity of NaOH solution isx,then the value of 20x will be-

54. Based on the information given below, the isoelectric point (P*I*) of lysine is _____________.



ANSWER KEY & SOLUTION

PART-I : PHYSICS

1.	B	**2.**	A	**3.**	B	**4.**	A	**5.**	C	**6.**	A	**7.**	ABCD
8.	AB	**9.**	CD	**10.**	BC	**11.**	BC	**12.**	D	**13.**	3.75	**14.**	2.17
15.	0.89	**16.**	0.75	**17.**	2.50	**18.**	6.28						

PART-II : MATHEMATICS

19.	D	**20.**	B	**21.**	A	**22.**	C	**23.**	D	**24.**	B	**25.**	AD
26.	ABCD	**27.**	AC	**28.**	ABC	**29.**	AB	**30.**	ABCD	**31.**	2.00	**32.**	2021.00
33.	12.00	**34.**	132.00	**35.**	2.00	**36.**	52.00						

PART-III : CHEMISTRY

37.	C	**38.**	D	**39.**	C	**40.**	C	**41.**	A	**42.**	C	**43.**	BD
44.	BCD	**45.**	ABD	**46.**	AB	**47.**	ACD	**48.**	ABCD	**49.**	250.00	**50.**	4.00
51.	8.00	**52.**	81.00	**53.**	2.00	**54.**	9.80						

SOLUTION | PART-I : PHYSICS | ADVANCED PAPER

1. **(B)**

2. **(A)**

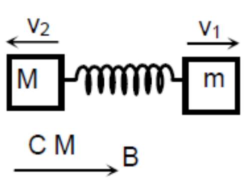
3. **(B)**

4. **(A)**

5. **(C)**

6. **(A)**

Sol.



7. **(A,B,C,D)**

Sol. Here the position y on the screen will correspond to maxima

$$y = \frac{n\lambda D}{d}$$

$$\lambda = \frac{30,000}{n} \text{ when } n = 1, 2, 3, 4, \ldots\ldots\ldots\ldots\ldots$$

8. **(A,B)**

Sol. $a = kx^2$

$\frac{da}{dx} = 2kx \Rightarrow \tan 60° = 2k\sqrt{3} \Rightarrow k = \frac{1}{2}$

$a = \frac{x^2}{2}$

$v\frac{dv}{dx} = \frac{x^2}{2} \Rightarrow \frac{v^2}{2} = \frac{x^3}{6} + C_1$ at $x = 0$, $v = 3$ m/s

$C_1 = \frac{9}{2}$

$v^2 = \frac{x^3}{3} + 9$

hence $a = 1.5$ and $v = \sqrt{\sqrt{3}+9}$

9. **(C,D)**

Sol. Capillary height $h = \frac{2T\cos\theta}{r\rho g} \Rightarrow h = 59.6$ mm

here 59.6 mm is greater than the protruding part hence water will rise in the capillary of insufficient height 25 mm.

Now, $R = \frac{2T}{hrg} = 0.6$ mm

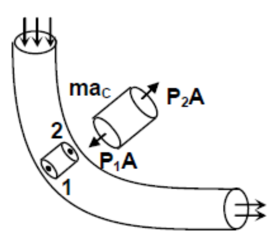
10. **(B,C)**

Sol. Fluid particles passing through the bend are in circular motion. The centripetal acceleration is provided by the variation in pressure. In the section as shown

$P_1A - P_2A = ma_c$

$\Rightarrow P_1 > P_2$

applying Bernoulli theorem at 1 and 2



i.e., $P_1 + \frac{1}{2}\rho_1 v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$

$P_1 + \frac{1}{2}\rho_1 v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$ as $h_1 = h_2$

$\because P_1 > P_2$

$v_1 < v_2$

11. **(B,C)**

Sol. $v_L = v_C = v_R;$

$\Rightarrow x_L = x_C = R$

when inductor is short circuited

$Z = \sqrt{R^2 + x_C^2} = \sqrt{2}R$

$\therefore I = \frac{30}{Z} = \frac{30}{\sqrt{2}R}$

$\therefore V_L = ix_L = \frac{30}{\sqrt{2}R} \times R = \frac{30}{\sqrt{2}}$

$\therefore$ (A) is incorrect and with similar calculations (B) will be correct.

Here f_0 is the resonance frequency as $v_L = v_C$

$\omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$

$\frac{x_L}{x_C} = \frac{\omega L}{1/\omega C} = \omega^2 LC$

Given $f = 2f_0$

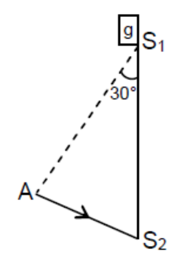
$\Rightarrow \quad \omega = 2\omega_0$

$\therefore \quad \frac{x_L}{x_C} = 4$

$\therefore$ (C) is also correct.

12. **(D)**

Sol. Extra phase change in glass = phae change in water of length AS_2.



$$\frac{2\pi t}{\lambda g} - \frac{2\pi}{\lambda\omega} t = \frac{2\pi}{\lambda\omega} . \frac{2}{3} \sin 30°$$

$$\Rightarrow t\left[\frac{3}{2} - \frac{4}{3}\right] = \frac{4}{3} \times \frac{1}{3}$$

$$\Rightarrow t = \frac{3}{8} \text{mm}.$$

13. **(3.75)**

Sol. $T = \frac{3}{8} M\omega^2 L^2$

$$\text{Stress} = \frac{F_\perp}{(A/\cos\theta)} = \frac{3}{8} \frac{M\omega^2 L^2 \cos^2\theta}{A}$$

14. **(2.17)**

Sol. $\frac{2.17 + 2.17 + 2.18}{3} = 2.1733\ldots\ldots \approx$ 2.17 mm upto 3 significant figure.

15. (0.89)

Sol. $2g - T = 2a$ .....(i)

$TR = I\alpha$ .....(ii)

$a = R\alpha$ .....(iii)

From (ii) and (iii) $T = \frac{Ia}{R^2}$

$\therefore\ 2g = a\left(2 + \frac{I}{R^2}\right)$

$\Rightarrow a = \frac{2g}{\left(2 + \frac{I}{R^2}\right)} = \frac{2 \times 9.8}{2 + \frac{0.2}{0.01}} = \frac{9.8}{11} \approx 0.89\ \mathrm{m/s^2}$

16. (0.75)

Sol. $(F_s)_{max}$ = 20 N. amax of 20 kg black = 1 m/s2

$\therefore$ both blck move together till 6 sec

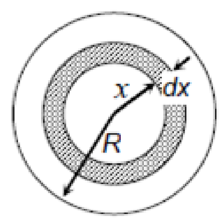
$\therefore\ \frac{dv}{dt} = \frac{5t}{30} \Rightarrow \int dv = \frac{1}{6}\frac{t^2}{2}$; $V = \frac{3^2}{12} = \frac{9}{12} = 0.75\mathrm{m/s}$

17. (2.50)

Sol. $d\phi = (2\pi x dx)kxt^2$

$\phi = \frac{2}{3}kt^2\pi x^3$

$\frac{d\phi}{dt} = \frac{4k\pi tx^3}{3}$



$E2\pi x = \frac{4\pi ktx^3}{3}$; $E = \frac{2}{3}ktx^2; d\tau = \left(\frac{2}{3}ktx^2\right)\frac{Q}{\pi R^2}(2\pi x dx)x$

$\int d\tau = \frac{4}{3}\frac{ktQ}{R^2}\int_0^R x^4 dx \quad \Rightarrow \tau = \frac{4}{3}\frac{ktQ}{R^2}\frac{R^5}{5}$

At $t = 15\mathrm{sec}, \tau = 2.50\mathrm{Nm}$

18. (6.28)

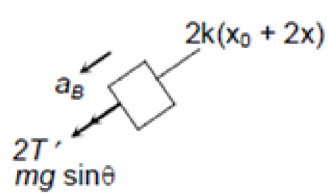
Sol. Let elongation of spring be x_0 in equilibrium. Then,

$2T + mg\sin\underline{\theta} = 2kx_0$ ....(i)

and $T = mg$ ......(ii)

Let Block B is displaced by x down the inclination

F.B.D. of B

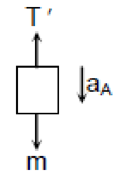


$-ma_B = 2K(X_0 + 2X) - 2T' - mg\sin\theta$ .....(iii)

F.B.D of A

$mg - T' = ma_A$

Also, $a_A = 2a_B$



$T' = mg - 2ma_B$

$-ma_B = 4kx_0 + 4kx - 2mg + 4ma_B - mg\sin\theta$

$a_B = -\frac{4k}{5m}x$

$\therefore\ T = 2\pi\sqrt{\frac{5m}{4k}}$

T = 6.28 s.

PART-II : MATHSMATICS

19. [D]

Sol. There are $^7C_5$ ways of selecting the rings to be worn. If a, b, c, d are the numbers of rings on the fingers, we need to find the number of quadruples (a, b, c, d) of non-negative integers such that a + b + c + d = 5. The number of such quadruples is

$$^{(5+4-1)}C_{(4-1)} = (^8C_3)$$

For each set of 5 rings, there are 5! assignments. Therefore, the total number of required arrangements is

$$^7C_5 \times {}^8C_3 \times 5! = 141120$$

20. [B]

Sol. Let $E_1 = 0$ and $E_2 = 0$ be the given two planes and $E = 0$ be the required plane. Since $E = 0$ passes through the line of intersection of $E_1 = 0$ and $E_2 = 0$,

$$E = E_1 + \lambda E_2 = 0$$

$\Rightarrow \quad E \equiv (x + 2y + z - 1) + \lambda(2x + y + 3z - 2) = 0$

$\Rightarrow \quad E \equiv (1 + 2\lambda)x + (2 + \lambda)y + (1 + 3\lambda)z - 1 - 2\lambda = 0$

This is perpendicular to the plane $x + y + z - 1 = 0$. So

$$1(1 + 2\lambda) + 1(2 + \lambda) + 1(1 + 3\lambda) = 0$$

$\Rightarrow \quad 6\lambda + 4 = 0$

$\Rightarrow \quad \lambda = \dfrac{-2}{3}$

Therefore

$$E \equiv \left(1 - \frac{4}{3}\right)x + \left(2 - \frac{2}{3}\right)y + (1 - 2)z - 1 + \frac{4}{3} = 0$$

$\Rightarrow \quad E \equiv -x + 4y - 3z + 1 = 0$

$\Rightarrow \quad E \equiv x - 4y + 3z - 1 = 0$

Hence, k = –4.

21. [A]

Sol. Differentiating the given function we get

$$f'(x) = 3(m + 2)x^2 - 6mx + 9m$$
$$= 3[(m + 2)x^2 - 2mx + 3m]$$

Since f is decreasing on R, we have $f'(x) \le 0 \ \forall \ x \in R$.

Therefore

$$(m + 2)x^2 - 2mx + 3m \le 0 \ \forall \ x \in R$$

So, $4m^2 - 12m(m + 2) \le 0$ and $m + 2 < 0$

$m^2 - 3m(m + 2) \le 0$ and $m < -2$

$-2m(m + 3) \le 0$ and $m < -2$

$m(m + 3) \ge 0$ and $m < -2$

$m \le -3$ or $m > 0$ and $m < -2$

$m \le -3$ and $m \ge 0$

But, when m = –3 or m = 0, $f(x)$ is not decreasing for all $x \in R$ (check this point). Therefore

$$m < -3 \text{ or } m > 0$$

So, $\quad m \in (-\infty, -3) \cup (0, \infty)$

22. **[C]**

Sol. $C_2 : \frac{x^2}{9} - \frac{y^2}{7} = 1$

Let any point P(3secθ, $\sqrt{7}$ tanθ) on C_2

Image of P in line x + y = $\sqrt{2}$ is θ(h, k), then

$\frac{h - 3\sec\theta}{1} = \frac{k - \sqrt{7}\tan\theta}{1}$

$= \frac{-2(3\sec\theta + \sqrt{7}\tan\theta - \sqrt{2})}{1+1}$

⇒ h = $-\sqrt{7}$ tanθ + $\sqrt{2}$, k = –3 secθ + $\sqrt{2}$

⇒ Equation of curve C_1 is

$$\frac{(x-\sqrt{2})^2}{7} - \frac{(y-\sqrt{2})^2}{9} = 1$$

Now, given line is tangent to the curve C_2 at $\left(\frac{9}{\sqrt{2}}, -\frac{7}{\sqrt{2}}\right)$ and perpendicular to x + y = $\sqrt{2}$

i.e., x – y = $8\sqrt{2}$.

23. **[D]**

Sol. $P = 1 - \left[\frac{2}{10}\times\frac{8}{9}\times\frac{1}{8}\times\frac{4}{10}\times\left[\frac{3}{9}\times\frac{2}{8}+\frac{6}{9}\times\frac{3}{8}\right]\times 2\right]$

$= 1 - \left(\frac{16+48+144}{10.9.8}\right) = 1 - \frac{208}{10.9.8}$

$= 1 - \frac{26}{10.9} = 1 - \frac{13}{45} = \frac{32}{45}$

24. **[B]**

Sol. $(1 + {}^7C_1x^4 + \ldots\ldots)(1 + x^2)^3(1 - x)^3$

$(1 + {}^7C_1x^4)(1 + x^6 + 3x^4 + 3x^2)(1 - x^3 + 3x^2 - 3x)$

∴ coefficient of x^5

= – 9 – 3 – 21 =–33

25. **[A,D]**

Sol. $z^{11} - 1 = (z - 1)(z - z_1)(z - z_2) \ldots\ldots (z - z_{10})$

$\Rightarrow (z - z_1)(z - z_2) \ldots\ldots (z - z_{10}) = \frac{z^{11}-1}{z-1}$

At z = i

$\prod_{k=1}^{10}(i - z_k) = \frac{i^{11}-1}{i-1} = \frac{-i-1}{i-1} = i$

$(-1)^{10}\prod_{k=1}^{10}(\omega + z_k) = \frac{(-\omega)^{11}-1}{-\omega-1}$

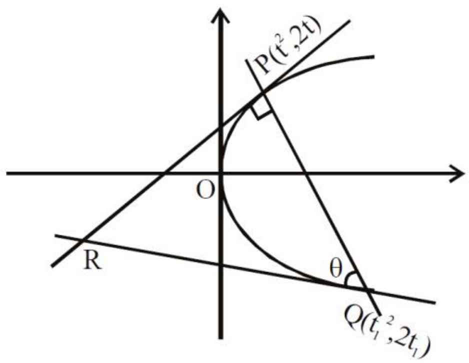
$\Rightarrow \prod_{k=1}^{10}(\omega + z_k) = \frac{-\omega^{11}-1}{-\omega-1} = \frac{-\omega^2-1}{-\omega-1} = \frac{\omega}{\omega^2} = \omega^2$

26. **[A,B,C,D]**

Sol. Let $P(x_0, y_0)$ be $(t^2, 2t)$

$\Rightarrow$ $x = t^2$, $y_0 = 2t$

$t_1 = -t - \frac{2}{t}$ $\Rightarrow tt_1 = -t^2 - 2$



$$\tan\theta = \left|\frac{-t - \frac{1}{t_1}}{1 + (-t).\frac{1}{t_1}}\right| = \left|\frac{tt_1 + 1}{t_1 - t}\right| = \frac{|t|}{2} = \frac{|y|}{4}$$

Length of normal chord

$$PQ = \sqrt{(t^2 - t_1^2)^2 + 4(t - t_1)^2}$$

$$= |t - t_1|\sqrt{(t + t_1)^2 + 4}$$

$$= \left|2t + \frac{2}{t}\right|\sqrt{\frac{4}{t^2} + 4}$$

$$= \frac{4(t^2 + 1)^{3/2}}{t^2} = \frac{4(x_0 + 1)^{3/2}}{x_0}$$

Coordinates of R are $(tt_1, t + t_1)$

$\Rightarrow$ ordinate of R is $t + t_1 = -\frac{2}{t} = -\frac{4}{y_0}$

$\because$ Triangle PQR is right angled.

$\therefore$ Its circumcentre is mid point of QR

$\Rightarrow$ Its ordinate is $\frac{t + t_1 + 2t_1}{2} = \frac{-2(x_0 + 3)}{y_0}$

27. **[A,C]**

Sol. Given equation

$\sec^2(x + y)dx + 2\sin x dx + \sec^2(x + y)dy + 2\sec^2 y\, dy = 0$

$\sec^2(x + y)(dx + dy) + 2\sin x dx + 2\sec^2 y dy = 0$

$\therefore$ $\tan(x + y) - 2\cos x + 2\tan y + c = 0$

$\because$ $y(0) = \frac{\pi}{4}$ $\Rightarrow 1 - 2 + 2 + c = 0$

$\Rightarrow$ $c = -1$

$\tan(x + y) - 2\cos x + 2\tan y - 1 = 0$

$\therefore \quad y\left(\frac{\pi}{2}\right) = \frac{\pi}{4} \qquad \left(\because y \in \left[0, \frac{\pi}{2}\right)\right)$

Put $\quad x = \frac{\pi}{2}, y = \frac{\pi}{4}$ in differential equation

$\Rightarrow \quad \frac{dy}{dx} = -\frac{2}{3}$

$\Rightarrow$ Equation of tangent to the curve at $\left(\frac{\pi}{2}, \frac{\pi}{4}\right)$ is $8x + 12y = 7\pi$

28. [A,B,C]

Sol. Area of $\Delta PAB = 100\sin\theta \cos^3\theta$

$\frac{3\sqrt{3}}{16}\sin\theta.\cos^3\theta$

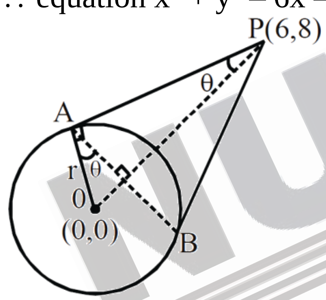
$\therefore \quad \theta = \frac{\pi}{6} \Rightarrow r = 5$

Equation of chord of contact is T = 0

$6x + 8y = r^2$

OP is diameter of circumcircle of ΔABP

$\therefore$ equation $x^2 + y^2 - 6x - 8y = 0$



29. [A,B]

Sol. $\lim_{x \to 0^+} f(x) = \lim_{x \to 0^-} f(x) = f(0)$

$$\lim_{x \to 0^+} \frac{\sqrt{x}[1 + rx]^{1/2} - \sqrt{x}}{r\sqrt{x}} = \lim_{x \to 0^-} \frac{\sin(p+q)x}{(p+q)x}(p+q) + \frac{\sin 2x}{2x}.2 = q$$

$\therefore \quad \frac{1}{2} = p + q + 2 = q$

$\Rightarrow \quad q = \frac{1}{2}, p = 2$

30. [A,B,C,D]

Sol. Let a be the first term and d be the common difference of the A.P. Let

$f(x) = 2x^2 - 4x + 10$

Therefore

$f'(x) = 4x - 4 = 0 \Leftrightarrow x = 1$

and $f''(x) = 4 > 0$

Therefore f has least value at x = 1 and the least value is

$f(1) = 2(1) - 4 + 10 = 8$

By the hypothesis,

$(a + 2d) + (a + 8d) = 8$

$a + 5d = 4$ …(1)

From 1,

$$d = \frac{4-a}{5}$$

Thus (A) is correct and also (B) and (C) are correct.

Now, the sum of the first eleven terms is

$$\frac{11}{2}[a + (a + 10d)] = 11(a + 5d)$$

$$= 11 \times 4 = 44$$

Thus (D) is correct.

31. **[2.00]**

Sol. Put z = iy in the equation

$-(\cos\theta + i\sin\theta)y^2 + iy + 1 = 0$ comparing

$$\left.\begin{array}{l} 1-\cos\theta.y^2 = 0 \\ 1 - y\sin\theta = 0 \end{array}\right\} \cap \text{ eliminating y, we get}$$

$\cos\theta = \sin^2\theta$

Now, $f(x) = x^3 - 3x^2 + 3(1 + \cos\theta)x + 5$

$f'(x) = 3x^2 - 6x + 3(1 + \cos\theta)$

$D = 36 - 36(1 + \cos\theta) = -36\cos\theta = -36\sin^2\theta \le 0$

$\therefore$ $f(x)$ is strictly increasing function and hence no local extrema also $f(0) = 5$ suggests that $f(x) = 0$ has one (–ve) real root.

$\therefore$ $p + q + r = 0 + 1 + 1 = 2$

32. **[2021.00]**

Sol. Given relation is

$$f(x) + f\left(\frac{1}{1-x}\right) = \frac{2}{x} - \frac{2}{1-x} \quad ....(1)$$

Replacing x with $\frac{1}{1-x}$ in Eq. (1) we have

$$f\left(\frac{1}{1-x}\right) + f\left(\frac{x-1}{x}\right) = 2(1-x) + \frac{2(1-x)}{x} \quad ....(2)$$

Again replacing x with $\frac{1}{1-x}$ in eq. (1), we get

$$f\left(\frac{x-1}{x}\right) + f(x) = -2x - \frac{2x}{1-x} \quad ....(3)$$

Now adding equation (1) and (3) and subtracting equation (2) gives

$$2f(x) = \left(\frac{2}{x} - \frac{2}{1-x} - 2x - \frac{2x}{1-x}\right) - 2(1-x) - \frac{2(1-x)}{x}$$

$$= \left(\frac{2}{x} - \frac{2(1-x)}{x}\right) + \left(\frac{-2}{1-x} - \frac{2x}{1-x}\right) - 2x - 2(1-x)$$

$$= \frac{2x}{x} - \frac{2(1+x)}{(1-x)} - 2$$

$$= 2 + \frac{2(x+1)}{x-1} - 2$$

$$= \frac{2(x+1)}{x-1}$$

Therefore

$$f(x) = \frac{x+1}{x-1}$$

33. **[12.00]**

Sol. Let A $\in$ S. In a symmetric matrix the (i, j)th element is same as (j, i)th element for i $\neq$ j. That is, upper and lower parts of the principal diagonal are reflections of each other through the principal diagonal. Hence the principal diagonal of A must have three 1's or two 0's and a single because A has five 1's and four 0's. If all the three diagonal elements are 1, the number of such matrices is $^3C_1$. If two diagonal elements are zeros and one is 1, then the number of such matrices is $^3C_1 \times {}^3C_1$. Therefore, total number of matrices in S = $^3C_1 + {}^3C_1 \times {}^3C_1 = 3 + 3 \times 3 = 12$.

34. **[132.00]**

Sol. Any point on the curve is

4(1 + cosθ), 3(1 + sinθ). Then find reflection of this point about the line and then find locus of that point.

35. **[2.00]**

Sol. Do yourself

36. **[52.00]**

Sol. Normal vector of required plane is parallel to vector

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -3 \\ 4 & 3 & 5 \end{vmatrix} = 14\hat{i} - 27\hat{j} + 5\hat{k}$$

$\therefore$ Equation of required plane is

$14(x - 1) - 27(y - 2) + 5(z - 3) = 0$

$\Rightarrow 14x - 27y + 5z + 25 = 0$

PART-III : CHEMISTRY

37. **(C)**

38. **(D)**

39. **(C)**

40. **(C)**

41. **(A)**

42. **(C)**

43. **(B,D)**

44. **(B,C,D)**

45. **(A,B,D)**

46. **(A,B)**

47. **(ACD)**

48. **(ABCD)**

49. **(250.00)**

50. **(4.00)**

Sol. Let x is the mole fraction of N_2H_4 in original mixture

amount of gases to start with = 1 mol

Amount of gases at end = [3x + 2(1 – x)] mol

$$\frac{P_1}{P_2} = \frac{n_1T_1}{n_2T_2}$$

$$\frac{0.5}{4.5} = \left(\frac{1}{2+x}\right)\left(\frac{300}{1200}\right)$$

x = 0.25

1/x = 4

51. **(8.00)**

52. **(81.00)**

53. **(2.00)**

54. **(9.80)**

QUESTION PAPER FORMAT AND MARKING SCHEME

19. The question paper has three parts: Physics, Mathematics and Chemistry.
20. Each part has three sections as detailed in the following table:

NAME OF THE CANDIDATE :____

ROLL NO. :______________________

I have read all the instructions and shall abide by them ______________________ Signature of the Candidate	I have verified the identity, name and roll number of the candidate. ___________________________ Signature of the invigilator